

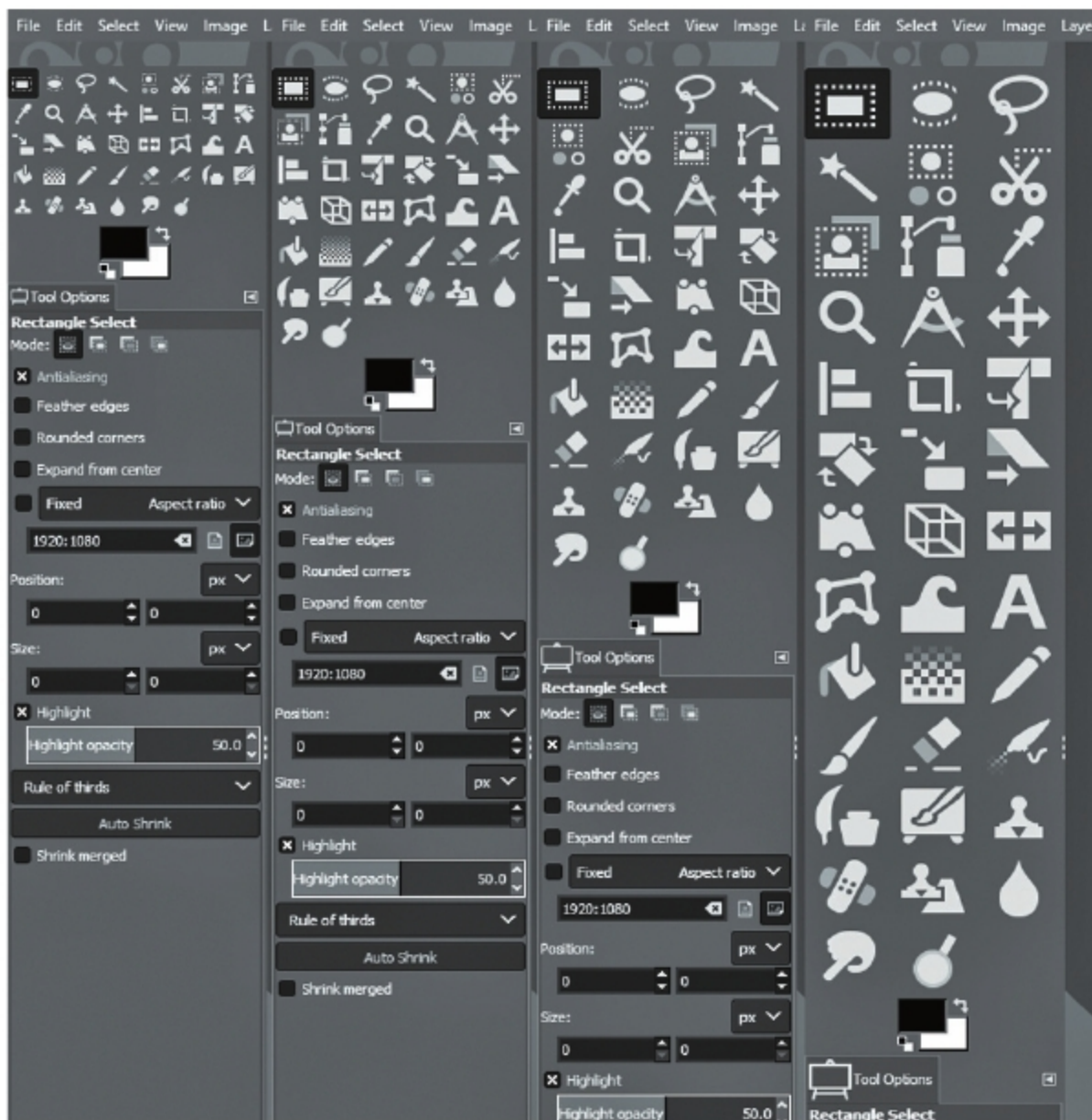


Figure 3: Icons in various sizes to adapt for HiDPI displays

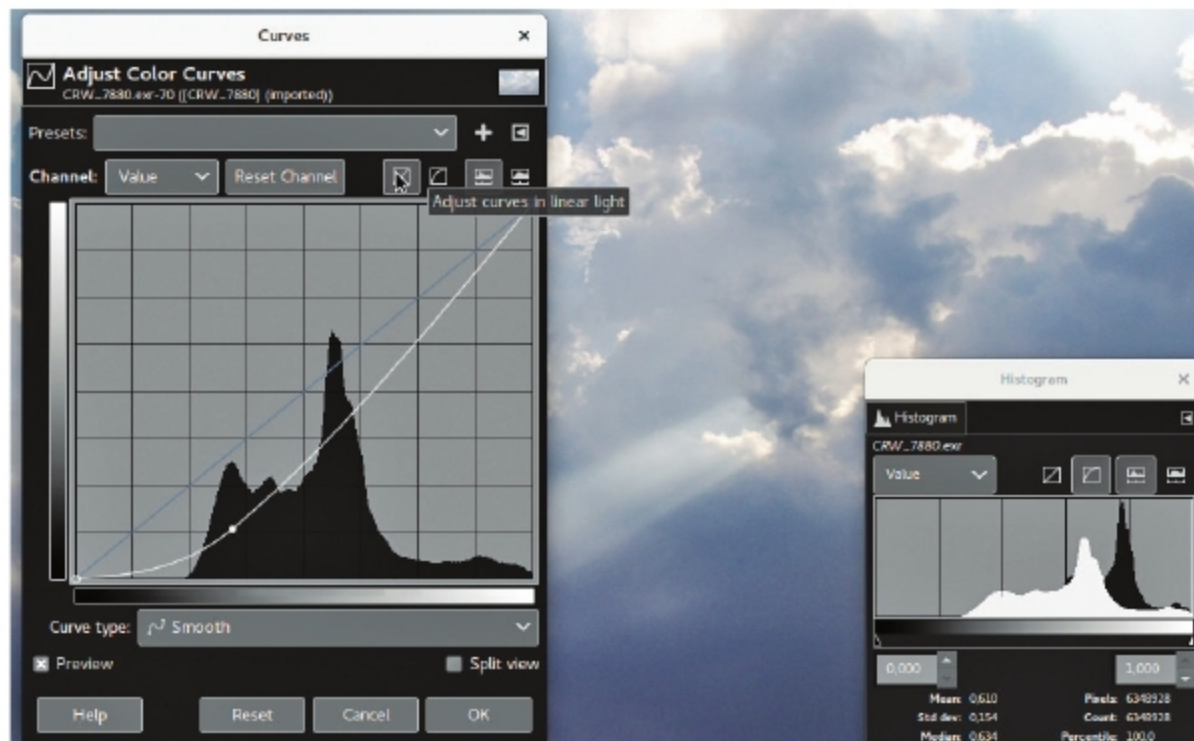


Figure 4: Adjust curves in linear light

time consuming or impossible. These include:

- *Clone tool*, which copies pixels using a brush
- *Healing brush*, which copies pixels

from an area and corrects the tone and colour

- *Perspective clone tool*, which works like the clone tool but corrects for distance changes

- *Blur and sharpen tool*, which blurs and sharpens images using a brush
- *Smudge tool*, which can be used to subtly smear a selection
- *Dodge and burn tool*, which is a brush that makes target pixels lighter (dodges) or darker (burns)

Layers, layer masks and channels

An image being edited in the GIMP can consist of many layers in a stack. The user manual suggests that, “A good way to visualise a GIMP image is as a stack of transparencies.” In GIMP terminology, each level (analogous to a transparency) is called a layer. Each layer in an image is made up of several channels. In an RGB image, there are normally three or four channels, each consisting of a red, green and blue channel. Colour sub-layers look like slightly different grey images, but when put together they make a complete image. The fourth channel that may be part of a layer is the alpha channel (or layer mask). It measures opacity, where a whole or part of an image can be completely visible, partially visible or invisible. Each layer has a layer mode that can be set to change the colours in the image.

There is support for several methods of sharpening and blurring images, including the ‘blur’ and ‘sharpen’ tools. The ‘unsharp mask’ tool is used to sharpen an image selectively — it sharpens only those areas of an image that are sufficiently detailed. This tool is believed to give more targeted results for photographs than a normal sharpening filter. The ‘selective Gaussian blur’ tool works in a similar way, except that it blurs areas of an image with little detail.

The latest version is GIMP 2.10, which is the result of six years of work that originally focused on porting the program to a new image processing engine, GEGL. However, the new version ships with far more new features, including new and improved tools, better support for file formats, various usability improvements, revamped colour management support, a plethora of